Daily Delve: Game Design Document

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Overview

Daily Delve is a daily roguelike deckbuilder in which every player faces the same seeded dungeon, and the fun is finding the card and charm combination that breaks *today's* run.

At midnight UTC a new seed produces the map, the events at each location, the enemies, the boss, the shops, and the pool of cards and charms in play. Players fight with cards, buy passive charms from merchants along the path they choose, fuse cards at forges, and compare scores on a global daily leaderboard. Runs take 15 to 25 minutes.

Design pillars

- **One dungeon, everyone.** Shared seeds make strategy shareable, so "reroll once at the second shop" is real advice.
- **Discovery over grind.** The daily card pool changes which engines are strong, so yesterday's winning build may fail today.
- **Mechanics are deterministic; flavor is generated.** An AI writes names and text. It never invents rules.
- **Choices have costs.** Charms, events, and fusions trade power for risk, so no pick is free.
- **Simple art, deep systems.** Vector shapes, pixel sprites, and low-poly Blender renders carry the look.

Target audience

Fans of Slay the Spire and Balatro who like short sessions, daily puzzles (Wordle-style habit), and theory-crafting builds with friends.

What makes it different

Feature	Why it matters
Shared daily seed	Fair leaderboards, replay sharing, community strategy

Feature	Why it matters
Bot-verified dungeons	Every published day is tested to be winnable and not trivially broken
Charm system	Run-level strategy (shops, routing, risk) on top of turn-level card play
Forge fusions with a daily secret recipe	A built-in reason to discover and share

Core loop and the daily seed

The whole game is a pure function: the same seed plus the same list of player choices always produces the same result.

Daily loop

1. Player opens the game and sees today's dungeon name, theme, modifiers, and map.
2. Player plans a route through the node graph (fights, elites, events, forges, shrines, merchants, bosses).
3. At each node the player picks from 2 or 3 seeded actions.
4. Fights use a deck of cards drawn from today's pool; rewards add cards, charms, relics, and gold.
5. The run ends at the final boss or at death. Score is submitted to the daily leaderboard.
6. Player watches ghost replays of top runs and reads the next day's Dungeon Gazette.

Run loop

1. Choose a path on the map.
2. Resolve the node (fight, event, shop, forge).
3. Gain or lose cards, charms, gold, HP.
4. Repeat until the act boss, then continue to the next act.

The determinism rule

Every random outcome is drawn from a named, seeded stream (map, enemies, loot, events, shop stock, rerolls, combat shuffle). A stream is only advanced by its own system, so a change in one place (say, an extra reroll) never shifts outcomes elsewhere. This gives

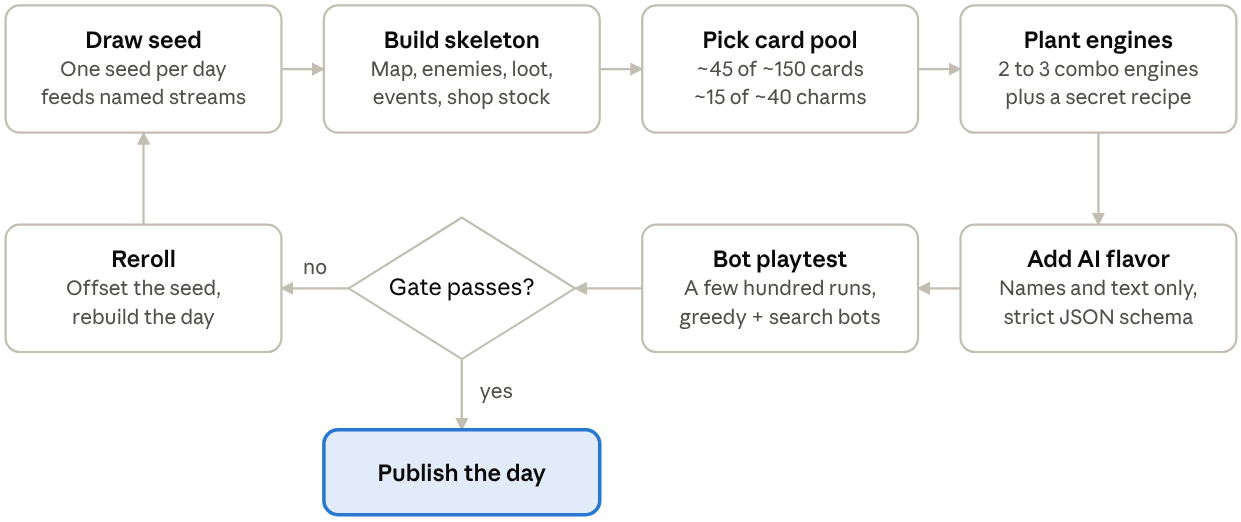
US:

- Replays that are just a seed plus a list of choices.
- Server-side cheat detection by re-simulating any submitted run.
- Fair leaderboards and shareable "reroll once here" tips.
- Bots that can play the same dungeon as humans.

Daily generation pipeline

A nightly job turns one seed into a published, bot-tested dungeon, and rerolls the day if the tests fail.

Only a bot-tested day gets published



nightly generation · 6 steps, 1 gate

A day is published only when the bots win between 15 and 50% of runs, at least one combo engine is completed in some runs, and no single charm pushes the win rate near 100%.

Pipeline rules

- **Separate streams.** Map, enemies, loot, events, and shop stock each draw from their own seeded stream, so changing one never shifts another.
- **Guaranteed engines.** The generator forces 2 to 3 combo engines into the day's pool, plus at least one charm that pairs with an engine, so a synergy is always discoverable.

- **Secret recipe.** One forge recipe per day is hidden from the codex.
- **AI flavor layer.** A model writes the dungeon name, theme, location names, event text, and boss personality into a strict JSON schema. It fills in text and picks from allowed templates. It never creates mechanics.
- **Reroll on failure.** A failed gate rerolls the whole day with a new seed offset, and the offset is recorded so the day stays reproducible.

Map and nodes

The map is a branching node graph in three acts of about six nodes each, so route planning is part of the daily puzzle.

Players see the full graph up front (node types may be hidden by some charms and modifiers), pick a path, and resolve one node at a time. Each act ends in a boss with a unique mechanic.

Node types

Node	What happens	Typical reward
Fight	Standard encounter	Gold, card pick
Elite	Harder fight with a special trait	Relic, better card pick
Event	Seeded choice, often a gamble	Varies, can cost HP or gold
Forge	Fuse two cards into one	A new combined card
Shrine	Remove a card, upgrade a card, or heal	Deck or HP improvement
Merchant	Buy cards, charms, relics; reroll stock	Build-defining purchases
Boss	End of each act, unique mechanic	Big reward, act transition

Seeded actions per node

Each node offers 2 or 3 actions fixed by the seed. Examples for a fight node:

- **Ambush:** start the fight with a free card in play.
- **Fight fair:** no modifiers.
- **Sneak past:** skip the fight, lose some gold.

Because actions and node placement are known in advance, planning a route around merchants, forges, and shrines is a real decision. A path with a Merchant is a path with a chance at a charm.

Combat and cards

Combat is turn-based with visible enemy intents: draw 5 cards, gain 3 energy, play cards, then enemies act.

Turn structure

1. Start of turn: draw 5 cards, energy resets to 3, start-of-turn effects trigger.
2. Player phase: play cards by paying energy; end turn when ready.
3. Enemy phase: enemies execute the intents shown at the start of the turn.
4. End of turn: unplayed cards are discarded, status effects tick.

Card types

Type	Behavior
Attack	Deals damage
Skill	Shield, draw, utility
Power	Permanent effect for the rest of the fight
Curse	Harmful card added by events, charms, or enemies

Tags

Every card carries one or more tags, which are the foundation of synergies: Fire, Frost, Poison, Summon, Shield, Echo, Blood, Arcane.

Enemies and relics

- Enemies telegraph intents (attack, block, buff, debuff) so planning is about the player's decisions, not guessing.
- Elites and bosses carry traits that favor or punish particular engines, which is how a day's boss interacts with the day's card pool.

- Relics are passive rewards from elites and bosses. They are distinct from charms, which are bought at merchants.

Combo system

Combos are discoverable at three depths, so newcomers find easy synergies on day one and veterans chase engines that need three or more pieces.

Layer 1: tag synergies

Cards react to tags played this turn. Example: "Deal 4 damage. +3 for each Fire card played this turn." These are visible in card text and easy to spot.

Layer 2: keyword chains (engines)

A small keyword language lets cards trigger each other.

Keyword	Effect
Burn / Chill / Poison	Status effects that tick or set up other effects
Echo	Replay the next card at half power
Cascade	When this kills, play the top card of your deck for free
Exhaust	Remove from the fight, powering up other cards
Shatter	Detonate Chill on an enemy for burst damage

Layer 3: forge fusions

At a Forge node the player combines two cards into one new card. Recipes come from tag pairs.

Ingredients	Result	Effect
Fire + Frost	Steam Burst	Damage to all enemies, ignores Shield
Poison + Echo	Lingering Plague	Poison replays each turn
Summon + Blood	Pact Familiar	Summon that grows when you take damage

Each day one **secret recipe** is seeded and left out of the in-game codex. Players who find it share the discovery, which feeds community engagement.

Example engine: Frost Lock

Chill an enemy with Frost cards, Shatter it with an Arcane card, and let Echo double the Shatter. Assembling all three in one turn can one-shot an elite.

Combo design rules

- Every day's card pool must contain at least two engines that are completable with 2 to 3 cards (see the generation pipeline).
- Combos should be readable: cards that interact with the card in hand are highlighted in the UI.
- No combo may be fully unbounded. Cap repeats per turn and cap scaling so one engine cannot trivialize every day.

Charm system

Charms are passive, run-long modifiers bought at Merchant nodes. Cards shape turn-to-turn play, and charms shape run-level strategy: how you shop, route, and take risks.

Rules

- **Slots:** start with 3, expandable by rare events or boss rewards up to 5.
- **Always active:** no energy or activation needed.
- **Permanent but sellable:** a charm stays for the run unless sold for roughly half price, which frees the slot.
- **Unique per run:** no duplicates, which prevents silly stacking.
- **Rarity:** Common, Uncommon, Rare, and Cursed (big upside, real downside, cheaper to buy).

- **Drawbacks scale with power:** each charm has a power score, and its cost must offset it.

Starter charm list

Shop and economy

Charm	Effect	Drawback
Window Dresser	Opening stock cards cost 1 coin	Rerolled cards cost +1
Haggler's Tongue	Rerolls cost 1 less	You can't sell cards
Magpie Ring	+25% gold from fights	Elites have +10% HP
Pawnbroker's Scale	Sell charms and relics for full price	Buying charms costs +2

Stat boosters

Charm	Effect	Drawback
Iron Locket	+8 max HP	Draw 4 instead of 5 on turn 1
Quick Fuse	+1 energy on turn 1	Shield doesn't carry over turns
Heavy Heart	+1 base damage on Attacks	First Skill each turn costs +1
Glass Charm	+20% damage dealt	+20% damage taken

Card and combo support

Charm	Effect	Drawback
Tag Magnet	Card rewards favor your most-played tag	Never offers off-tag cards
Echo Chamber	First Echo card each turn replays at full power	Echo cards add a Curse to your discard
Forge Apprentice	Forge fusions cost 0 coins	Fused cards have -1 to a number
Pocket Archive	Hand size +1	Deck minimum raised to 15 cards

Risk and reward (Cursed)

Charm	Effect	Drawback
Gambler's Tooth	Event gambles succeed far more often	Failed gambles cost double
Blood Ledger	Gain 1 gold when you take damage	Healing is halved
Hollow Crown	Start each act with a free rare card	Max HP -15%
The Tab	Buy anything free once per shop	Pay back double at the next shop

Map and route

Charm	Effect	Drawback
Cartographer's Eye	Reveals node types 2 steps ahead	Fights on revealed paths have +1 enemy
Lucky Rabbit's Foot	Events show outcome ranges up front	One fewer Merchant visible on the map

How charms create decisions

- **Route planning:** a shop-focused charm is strong on a map with three Merchants and weak on one.
- **Shop versus reroll tension:** Window Dresser makes the first look cheap, so rerolling has a real cost. Haggler's Tongue flips that.
- **Charm-card synergy:** Tag Magnet with a Fire deck snowballs, and Echo Chamber with Lingering Plague is a classic pairing.
- **Charm-charm combos:** Pawnbroker's Scale plus Window Dresser turns the economy into a weapon.
- **Uneven drawbacks:** Heavy Heart's cost barely matters for a cheap-Skill deck, so the same charm is a different choice for different builds.

Shop and reroll

Every Merchant sells a seeded opening stock, and rerolling replaces it with the next seeded set at an escalating price, so knowing when to reroll is part of the daily puzzle.

Stock composition

Slot	Count
Cards	5
Charms	2
Relic or consumable	1
Card removal service	1 (price rises each use)

Opening stock versus rerolls

- The first set shown at a Merchant is the **opening stock**. Everything after is a **reroll**. Charms can key off this distinction (for example Window Dresser).
- Reroll cost escalates: 2, 3, 4 coins and so on, per Merchant.
- The nth reroll at a given Merchant is the same for every player, so rerolls are shareable knowledge ("reroll once at the second shop, there's a Rare charm").
- Reroll results come from the shop stream only, so rerolling never changes combat or event outcomes.

Economy sinks and sources

- **Sources:** fight gold, elite gold, event rewards, selling cards and charms, and a few charms.
- **Sinks:** card prices, charm prices, rerolls, forge fees, and removal service.
- Charms that bend the economy (Window Dresser, Pawnbroker's Scale, The Tab) are priced higher and capped so they cannot dominate the leaderboard on their own.

Price guidance

Item	Rough price (coins)
Common card	40 to 60
Rare card	90 to 130
Common charm	30 to 50
Uncommon charm	55 to 80
Rare charm	90 to 130
Cursed charm	20 to 40

These are starting points for balance testing, not final numbers.

Daily modifiers

Each day carries one or two rule twists that shift which engines are strongest, so yesterday's winning build may fail today.

Modifier	Effect	Favors
Hot and Cold	Fire cards cost 1 less, Frost cards cost 1 more	Fire builds
Thin Blood	Healing is halved, start with a Blood relic	Blood and aggressive builds
Iron Hides	Enemies gain Shield each turn	Armor-piercing and poison builds
Tight Purse	Charm slots start at 2, charms cost 30% less	Cheap, high-impact charms
Cursed Bargain	Cursed charms are free, but you must take one	Players who can absorb a drawback
Long Shadows	Enemies show intents one turn late	Defensive, flexible decks

Rules

- A day gets at most two modifiers, and they may not directly cancel each other.
- The bot playtest gate runs with modifiers applied, so a modifier cannot make the day unwinnable.
- Modifiers are data entries with hooks, the same system charms use.

Scoring and social

The daily score rewards depth first, with bonuses for speed and style, and everything is verifiable because a run is just a seed plus choices.

Score

Component	Notes
Depth	Nodes cleared and bosses defeated, the main component
Speed bonus	Fewer turns across the run
Style bonus	Points per combo, fusion, and engine triggered, so risky wild plays can win
Gold conversion	Leftover gold converts to score at a capped rate so economy charms cannot dominate

Exact weights are set during balance testing.

Global leaderboard: who got farthest

The headline leaderboard ranks every player worldwide by how far they got in today's dungeon, with score only breaking ties.

Rank order	Rule
1. Furthest point reached	Act and node depth on the map; defeating the final boss counts as a full clear
2. Total score	Higher wins among players who reached the same point

Rank order	Rule
3. Fewest turns	Faster clears win remaining ties
4. Earliest submission	Final tiebreak

Each leaderboard row shows rank, player name, furthest node (for example "Act 3, Node 4" or "Cleared"), score, and a compact build summary (charms and key cards) with a replay button.

Views

- Global daily, friends, and per-modifier boards.
- All-time boards for most clears, longest streak, and highest single-day score.
- Daily boards reset at midnight UTC with the new seed, and past days stay browsable.

Graveyard map. After finishing, the player sees the day's map with a marker on each node showing how many players died there and a flag at the farthest node anyone reached. It doubles as a difficulty heat map and tells players which fights to prepare for.

Only the one official attempt per day counts for the global board. Practice runs never appear on it.

Attempts

- One official attempt per day, plus unlimited practice runs on the same seed.
- Streaks reward daily play.

Social features

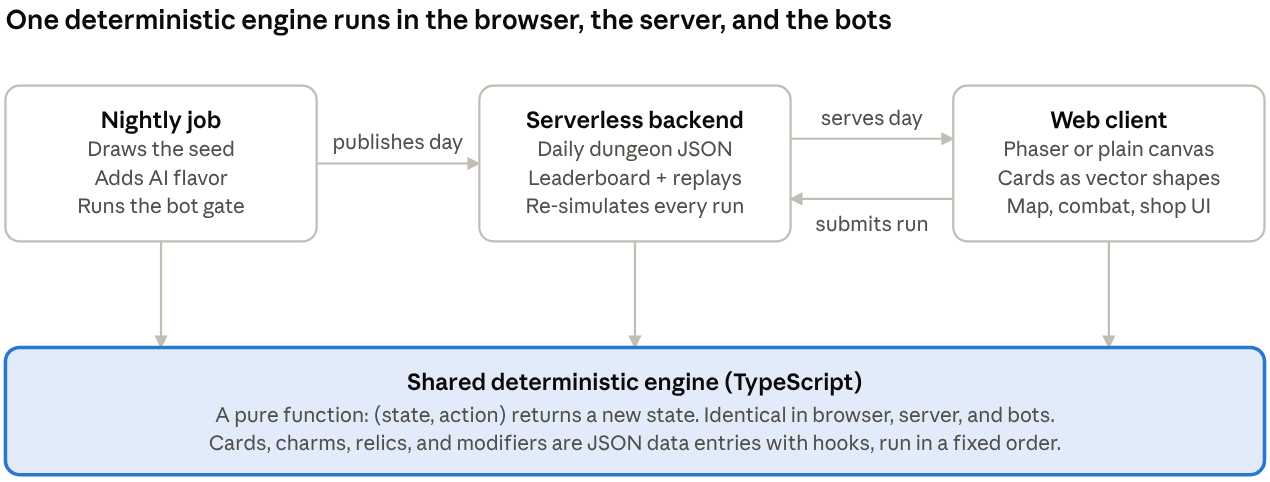
- **Replay codes:** a short code containing the seed and choice list, shareable in chat.
- **Ghost replays:** after finishing, watch how top players routed the dungeon.
- **Leaderboards:** global, friends, and per-modifier views.
- **Dungeon Gazette:** the next morning an AI writes a short recap of popular combos, notable runs, and who broke the dungeon, using real run statistics.
- **Secret recipe reveal:** the day's hidden fusion is revealed in the Gazette, crediting the first discoverer.

Integrity

The server re-simulates every submitted run from the seed and choices. A run whose result does not match is rejected, which makes cheating on the leaderboard impractical.

Technical architecture

The whole game rests on one pure, deterministic TypeScript engine that the client, the server, and the bots all run, so a run is only ever a seed plus a list of choices.



architecture · 3 services on one shared engine

Engine rules

- **Pure function:** `(state, action)` returns a new state, with no clock and no unseeded randomness.
- **Data, not code:** cards, charms, relics, and modifiers are JSON entries with trigger hooks and modifiers.
- **Fixed order of operations:** additive modifiers first, then multiplicative, then caps. Documented once, so combos behave predictably and bots cannot find loopholes.

Charm and modifier hooks

Hook	Fires when
onShopOpen	A Merchant node opens
onShopReroll	The player rerolls shop stock

Hook	Fires when
onFightStart	A combat begins
onCardPlayed	Any card is played
onDamageTaken	The player takes damage
onGoldGained	The player gains gold

Components

Component	Role
Web client	Phaser or plain canvas; cards are rectangles with icons and text; a charm bar sits under the HP display and glows when a charm triggers
Serverless backend	Holds the daily seed and dungeon JSON, the global leaderboard, and replay verification
Nightly job	Draws the seed, runs the AI flavor step and the bot playtests, then publishes
Replay verification	Re-simulates each submitted run from the seed and choices before it counts

Art direction

Art stays deliberately simple so the project can ship with assets generated by code and by Blender scripts.

Style

- **Cards and UI:** flat vector shapes (SVG or canvas), tag icons, and clean frames.
- **Enemies:** pixel art sprites, or chunky low-poly models rendered in Blender from a fixed angle into 2D sprites.
- **Backgrounds:** gradients, silhouettes, and a few parallax layers per act.
- **Effects:** simple particle bursts and screen shake for combos, so combos feel good without costly animation.

Asset list for a first playable

Asset	Count	Method
Card frames and tag icons	8 tags, 4 types	Vector, generated
Card art	60	Simple icons or small pixel pieces
Enemy sprites	30	Pixel art or Blender renders
Backgrounds	10	Gradients and silhouettes
Charm icons	25	Vector
UI (buttons, map nodes, HP and energy)	about 40	Vector

Readability rules

- Tag colors are consistent everywhere (for example Fire is orange, Frost is cyan).
- A charm glows briefly whenever it triggers.
- Cards that interact with the card in hand are highlighted.

Balance and playtesting

Balance rests on two defenses: a power budget on every card and charm, and a nightly bot gate that rejects broken days.

Power budgets

- Every card and charm gets a power score. Drawbacks must offset the score (a +1 energy charm needs a real cost, a small stat charm a mild one).
- Nothing scales without a cap: gold gain, damage multipliers, repeat triggers, and stacking effects all have hard ceilings.
- Economy items are priced to match their strength and kept at Uncommon or above.

Nightly bot gate

Before a day is published, bots play it a few hundred times.

Check	Target
Bot win rate	15 to 50%
Discoverable engine	At least one combo engine is completed by bots in some runs
Charm check	No single charm produces a near-100% win rate
Modifier check	Day stays winnable with modifiers applied

A day that fails is rerolled with a new seed offset. Bots include a greedy heuristic player and a search-based player, and some runs force charm purchases so charm balance is tested.

Human testing

- Friend playtests in the last week to check combo readability and fun, which bots cannot judge.
- Tracked after launch: win rate per day, most-bought charms, most-played cards, and fusion discovery rate, with weekly balance patches.

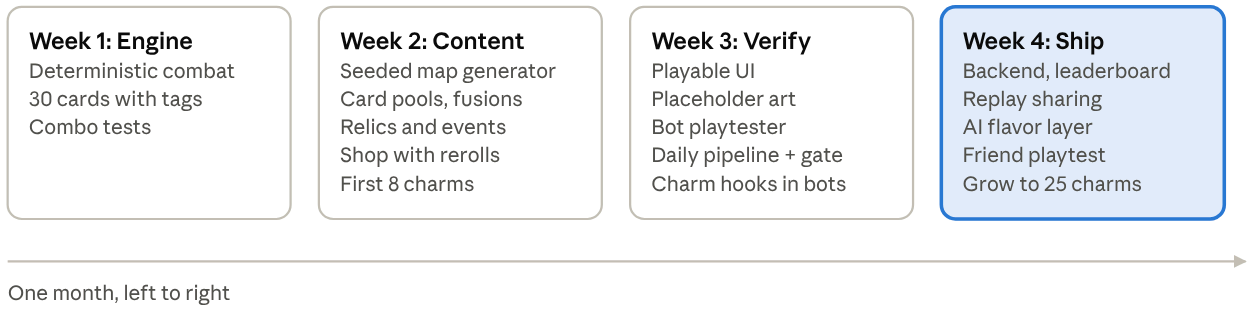
Guardrails

- Define one order of operations for modifiers (additive first, then multiplicative, then caps) and document it so combos behave predictably.
- Add content gradually (launch with about 60 cards and 25 charms) so each addition can be tested against the existing pool.

Roadmap

The plan is one month to a shipped daily dungeon, starting with the combat engine because that is where we learn whether the combos are actually fun.

Four weeks from engine to a shipped daily dungeon



roadmap · 4 weeks, 4 phases

Launch content scope

Content	Launch	Later
Cards	About 60	Grow toward 150, a few per week
Charms	About 25	Grow toward 40
Daily modifiers	A starter table of 6	New twists added over time
Daily pools	About 45 cards and 15 charms per day	Tuned by win-rate data

After launch

- Weekly balance patches driven by win rate per day, most-bought charms, and fusion discovery rate.
- New cards, charms, and modifiers added each week, each tested against the existing pool by the bot gate.
- Streak and leaderboard features tuned to keep the daily habit loop healthy.

Risks and open questions

The biggest risks are balance, combo readability, content volume, and daily retention.

Risks

Risk	Why it matters	Mitigation
Balance	Combos and charms can break a day	Power budgets, bot gate, weekly patches
Combo readability	Opaque synergies make players feel lost	Highlight interacting cards, glow on charm triggers, codex
Content volume	Around 150 cards is a lot for a small team	Launch with about 60 cards and 25 charms, add weekly
Retention	Daily games live or die on the habit loop	Streaks, leaderboards, Gazette, shareable replays
AI text quality	Flavor text can be off-tone or wrong	Strict JSON schema, template slots, review sample before publish
Cheating	Leaderboard trust	Server re-simulation of every run

Open questions

- [ ] How punishing should Cursed charms be: mild annoyance or run-defining?
- [ ] Should charms be sellable, given the extra balance edge cases?
- [ ] Should slot count be a resource (expandable) or fixed at 3?
- [ ] Web only at launch, or also mobile?
- [ ] Is the AI flavor layer a runtime call or pre-generated in the nightly job only?
- [ ] What is the monetization path: cosmetics, supporter pass, or seasonal content?
- [ ] Final score weights for depth, speed, and style.